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SONIC-K1: The First Open-Weights Universal Motion Tracker for the Booster K1, Trained on a Single GPU

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01 The idea: SONIC for a small humanoid

NVIDIA SONIC showed that a single policy can track almost any human motion, but it was built for the larger Unitree G1 and trained on 128 GPUs. We asked whether a lab with one GPU can have the same kind of tracker on a small robot. So we ported SONIC to the Booster K1, trained it from scratch on one GPU, fixed it for the real robot, and released everything.

	SONIC · Unitree G1 as reported	SONIC-K1 · Booster K1 ours
Held-out success	98.5%	**97.4%**
PHUMA success	97.2%	**93.6%**
Training data	700 h mocap	**9,991 clips**

G1 values as reported by SONIC; robot, data and test clips differ.

What we contribute

- The first open-weights general motion tracker for the Booster K1, to our knowledge.
- A full SONIC training run on one GPU, under 1% of SONIC's reported compute.
- Tested in Isaac Lab, in MuJoCo and on the real K1, with a short post-training for the robot's real torque limits and delay.
- Code, weights, robot app and evaluation tools, all public.

Training time, drawn to scale

SONIC — 128 GPUs × 7 days = 21,000 GPU-h

SONIC-K1 — 1 GPU = ~145 GPU-h

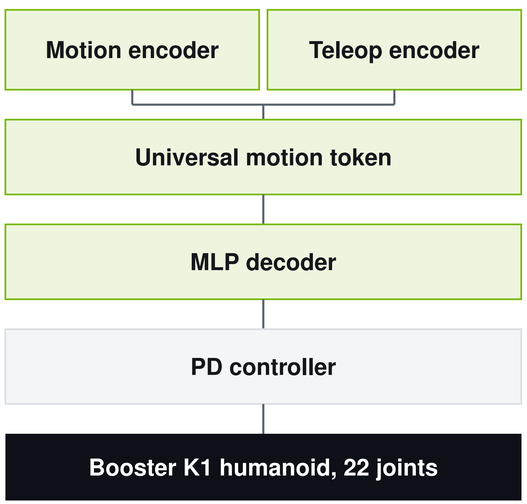
135.4 h of base training on one NVIDIA H100, plus 9.2 h of post-training on one RTX PRO 6000.

02 What we built: SONIC on the K1

SONIC is one network that tracks any reference motion. We kept its design, PPO settings, motion sampling and rewards, trained both encoders from scratch on K1 motion, and added a head-joint reward.

We retargeted 142,220 Bones-SEED human clips to the K1 with UMR and kept the 62,427 that the robot can physically perform, training on 9,991 and holding out 1,985 for testing. 16,384 simulated K1s then learn in parallel in Isaac Lab with GPU PhysX, all on one NVIDIA H100. A short post-training on one RTX PRO 6000 adds the robot's real torque limits and a random delay, and the final policy is exported to ONNX and runs onboard the K1 at 50 Hz. Simulation and training run end to end on the NVIDIA stack.

The SONIC network on the K1



Green = NVIDIA tools (the SONIC network)

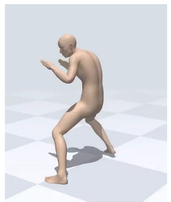
Post-trained: a short extra run with the K1's real torque limits and a random delay.

Built with NVIDIA

- **NVIDIA SONIC** network and recipe
- **GR00T-WBC** training code
- **Isaac Lab** RL training
- **Isaac Sim + PhysX** GPU simulation
- **NVIDIA H100** base training
- **RTX PRO 6000** post-training
- **SOMA** human body model

03 Tested in simulation and on the real K1

Every test starts from a human motion the policy has never seen, stored as a SOMA body (left). We retarget it to the K1 and play it in MuJoCo, a different physics engine from training, and then on the real robot (right).



Human reference, SOMA body



SONIC-K1, same motion

97.4%
of 1,985 held-out motions tracked

96.6% from 76.7%
post-trained, torque limits + 40 ms delay

A clip counts if root, hands and feet stay within 0.25 m of the reference height (SONIC's rule).

Bones-SEED held-out	97.4
100STYLE	99.0
PHUMA	93.6

Conclusion

A universal humanoid motion tracker can be ported to a small humanoid on a single GPU, without changing its network: SONIC-K1 tracks 97.4% of held-out motions with under 1% of SONIC's compute, and a short post-training with the robot's real limits makes it hold up under delay and run on the real K1.

Future work

Tracking error on the real K1 with motion capture, whole-body teleoperation with the trained teleop encoder, driving the robot from video, a larger model on all 62,427 feasible clips, and other small humanoids.

Everything is open

Code, robot app, eval tools · Apache-2.0 · github.com/Kyk0/sonic-k1
Weights v1.0 + v0.1, ONNX · CC BY-NC 4.0 · huggingface.co/Kyry10/sonic-k1
K1 USD asset, test-split list, laptop demo · kyry.me/sonic-k1

Limits and related work

Mostly simulation, one training seed, post-training changes not ablated. Any2Any and TRACE transfer G1 SONIC to full-size robots; we train from scratch for a smaller one.

Thanks

NVIDIA GEAR, Bones Studio, the UMR authors and Booster Robotics. Training data includes Motion Data by Bones Studio.